

Improving Male Sling Selectivity and Outcomes—A Potential Role for Physical Demonstration of Stress Urinary Incontinence Severity?

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Abstract

Introduction: We reviewed our 9-year experience with AdVance™ Male Sling System cases to determine clinical features associated with treatment success and to refine procedure selectivity. We hypothesized that preoperative physical demonstration of stress urinary incontinence by the standing cough test improves patient selection for male sling surgery.

Methods: Retrospective review of primary AdVance sling surgeries between 2008 and 2016 was performed. Patients without standing cough test results were excluded from study. Success was defined as 1 pad per day or less postoperatively and no further intervention. Standing cough test was performed during preoperative consultation and objectively graded using the MSIGS (Male Stress Incontinence Grading Scale).

Results: Of the 203 male patients who underwent sling placement 80 (39%) experienced treatment failure during a median followup of 63.5 months. From 2008 to 2016 the proportion of AdVance slings performed as a surgical treatment modality for stress urinary incontinence decreased from 66% to 13%. Increasing selectivity correlated with greater treatment success. Success was greater among men using 2 pads per day or less preoperatively (77% vs 36%, $p < 0.0001$), having physical findings of mild stress urinary incontinence (MSIGS grade 0-2 on standing cough test, 67% vs 26%, $p < 0.0001$) and without a history of radiation (64% vs 41%, $p = 0.02$). In combination, men without prior radiation with mild stress urinary incontinence and favorable standing cough test were “ideal patients” with an 81% success rate. Incremental increases in pad per day use (OR 1.8 per pad, $p < 0.0001$) and MSIGS grade (OR 1.7 per grade, $p = 0.005$) were independently associated with treatment failure.

Conclusions: Increasing selectivity has improved sling outcomes for men with stress urinary incontinence. Ideal sling candidates have not received radiation therapy, and have history and physical findings suggestive of mild stress urinary incontinence.

Abbreviations and Acronyms

AUS = artificial urinary sphincter
PGI-I = Patient Global Impression of Improvement
PPD = pad per day
SCT = standing cough test
SUI = stress urinary incontinence

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